SECSR2043 OPERATING SYSTEMS
[20 Marks]

Name	:	LAU YEE WEN	Marks
Student ID	:	A23CS0099	
Section	:	02	

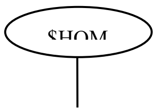
Instruction: Please answer all of the following questions. Whenever the ☝ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname*.sh (Example: `ahmad.sh`).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar .
```

a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **`$HOME`** directory.

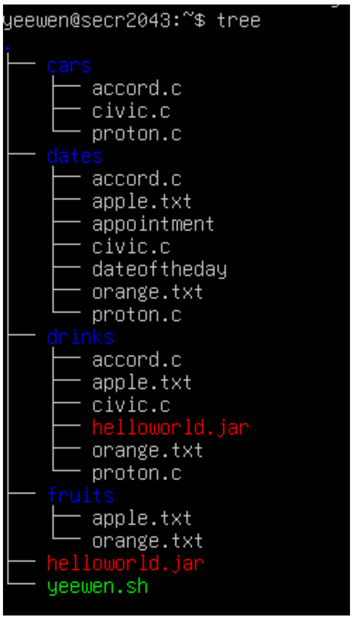
[4 marks]



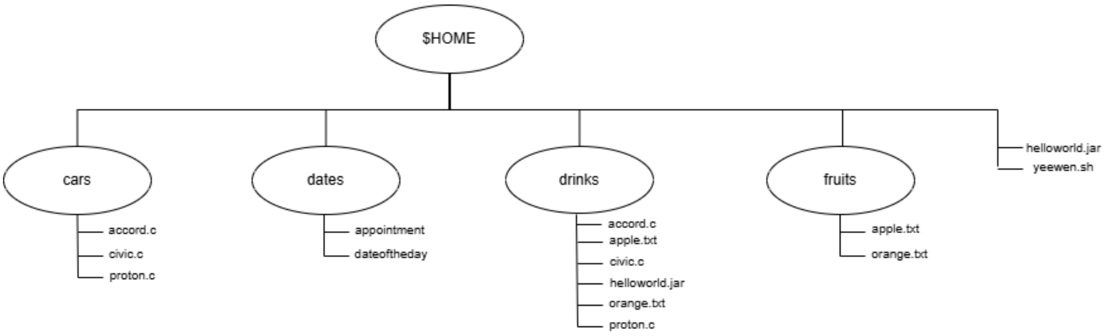
Print screen the script that you type;

```
GNU nano 7.2                                                     yeewen.sh
echo "Hello World" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Aircond" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd .. /fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.* .; cp ../fruits/*.* .;
cp ../*.jar.
```

Then draw the tree



```
yeewen@secr2043:~$ tree
.
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── accord.c
│   ├── apple.txt
│   ├── appointment
│   ├── civic.c
│   ├── dateoftheday
│   ├── orange.txt
│   └── proton.c
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── helloworld.jar
└── yeewen.sh
```



b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter "c" as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
 GNU nano 7.2                                                              ifiles.sh
#!/bin/bash

# Prompt the user to enter a file extension
read -p "Enter the file extension (without the dot):

# Find and display all files with the given extension
echo "Files with .${ext} extension:"
find . -type f -name "*.${ext}"

# Count the number of files with the given extension
count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Number of files with .${ext} extension: $count"
```

```
yeewen@secr2043:~$ nano ifiles.sh_
```

```
yeewen@secr2043:~$ chmod u+x ifiles.sh
yeewen@secr2043:~$ ./ifiles.sh
Enter the file extension (without the dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 6
yeewen@secr2043:~$ _
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

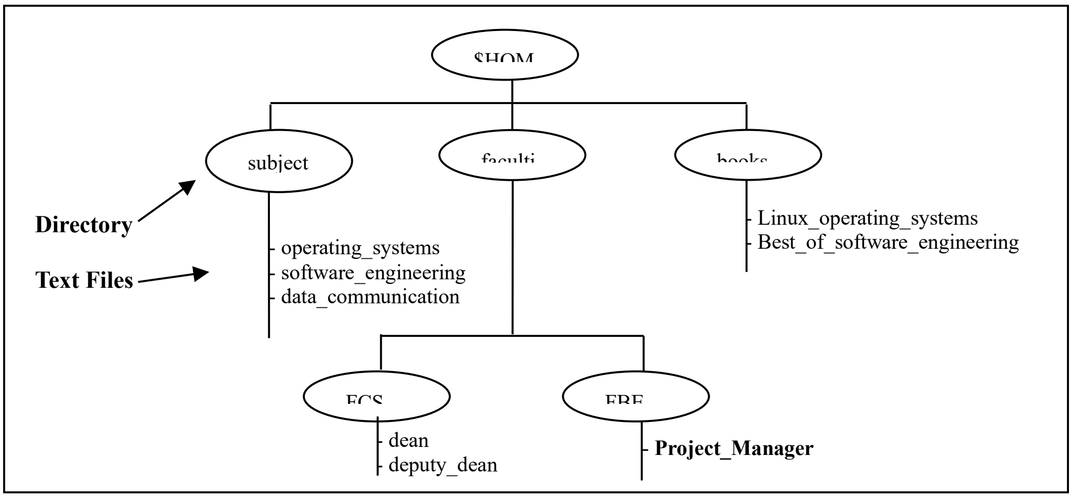


Figure 1

a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2                                                    yeewen2a.sh *
#!/bin/bash

#Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

#Create and write to text files in the subjects directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

# Create and write to text files in the faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

#Create and  write to text files in the  faculties/FBE directory
echo "Project Manager"  > $HOME/faculties/FBE/Project_Manager

#Create and write to text files in the  books directory
echo "Linux operating systems" >$HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

```
yeewen@secr2043:~$ nano yeewen2a.sh_
```

```
yeewen@secr2043:~$ chmod u+x yeewen2a.sh
yeewen@secr2043:~$ ./yeewen2a.sh
yeewen@secr2043:~$ tree $HOME
/home/yeewen
├── books
│   ├── Best_of_software_engineering
│   └── Linux_operating_systems
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── faculties
│   ├── FBE
│   │   └── Project_Manager
│   └── FCS
│       ├── dean
│       └── deputy_dean
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── helloworld.jar
├── ifiles.sh
├── subjects
│   ├── data_communication
│   ├── operating_systems
│   └── software_engineering
├── yeewen.sh
└── yeewen2a.sh

10 directories, 25 files
```

b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`. [2 marks]

Directory/File	Access Control
`books`	`drwxrwxr-x`
`subjects`	`drwxrwxr-x`
`Best_of_software_engineering`	`-rw-rw-r--`
`FCS`	`drwxrwxr-x`
`project_manager`	`-rw-rw-r--`

```
yeewen@secr2043:~$ ls -ld books
drwxrwxr-x 2 yeewen yeewen 4096 Jun 20 15:51 books
yeewen@secr2043:~$ ls -ld subjects
drwxrwxr-x 2 yeewen yeewen 4096 Jun 20 15:51 subjects
yeewen@secr2043:~$ ls -l books/Best_of_software_engineering
-rw-rw-r-- 1 yeewen yeewen 29 Jun 20 15:58 books/Best_of_software_engineering
yeewen@secr2043:~$ ls -ld faculties/FCS
drwxrwxr-x 2 yeewen yeewen 4096 Jun 20 15:51 faculties/FCS
yeewen@secr2043:~$ ls -l faculties/FBE/Project_Manager
-rw-rw-r-- 1 yeewen yeewen 16 Jun 20 15:58 faculties/FBE/Project_Manager
```

c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information: [4 marks]

Directory/File	Users								
	Owner			Group			Public		
`subjects`	✓	✓	✓	✓	x	x	✓	x	x
`Best_of_software_engineering`	✓	x	✓	x	✓	x	x	x	x
`FCS`	✓	✓	x	x	x	x	✓	✓	✓
`project_manager`	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
GNU nano 7.2                                                        yeewen2c.sh
#!/bin/bash

#Set permission for subjects directory
chmod 744 $HOME/subjects

#Set permission for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

#Set permission for FCS directory
chmod 607 $HOME/faculties/FCS

#Set permission for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager
```

```
yeewen@secr2043:~$ nano yeewen2c.sh_
```

```
yeewen@secr2043:~$ chmod +x yeewen2c.sh
yeewen@secr2043:~$ ./yeewen2c.sh
yeewen@secr2043:~$ ls -ld ~/subjects
drwxr--r-- 2 yeewen yeewen 4096 Jun 20 15:51 /home/yeewen/subjects
yeewen@secr2043:~$ ls -ld ~/faculties/FCS
drw----rwx 2 yeewen yeewen 4096 Jun 20 15:51 /home/yeewen/faculties/FCS
yeewen@secr2043:~$ ls -l ~/books/Best_of_software_engineering
-r-x-w---- 1 yeewen yeewen 29 Jun 20 15:58 /home/yeewen/books/Best_of_software_engineering
yeewen@secr2043:~$ ls -l ~/faculties/FBE/Project_Manager
-----wx--x 1 yeewen yeewen 16 Jun 20 15:58 /home/yeewen/faculties/FBE/Project_Manager
```

d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	drw----rwx
FCS	-r-x-w----
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****